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Energy prices and household heterogeneity: Monetary policy in a Gas-TANK[☆]

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ABSTRACT

How does household heterogeneity affect the transmission of an energy price shock? What are the implications for monetary policy? We develop a small, open-economy TANK model that features labor and an energy import good as complementary production inputs (Gas-TANK). Given such complementarities, higher energy prices reduce the labor share of total income. Due to borrowing constraints, this translates into a drop in aggregate demand. Higher price flexibility insures firm profits from energy price shocks, further depressing labor income and demand. We illustrate how the transmission of shocks in a RANK versus a TANK depends on the degree of complementarity between energy and labor in production and the extent of price rigidities. Optimal monetary policy is less contractionary in a TANK and can even be expansionary when credit constraints are severe. Finally, we show that the contractionary effect of energy price shocks on demand cannot be generalized to alternate supply shocks.

1. Introduction

In early 2022, energy prices rose to historically high levels as Russia's invasion of Ukraine increased the risk of disruptions to the energy trade. For energy importers such as the UK or the EA, these developments represented a deterioration in the terms of trade, implying a contraction in income flowing to domestic production inputs, including labor income. If households face limits in their access to financial markets, the contraction in income can translate into a drop in aggregate demand. An energy price shock, typically modeled as a supply shock, can therefore have a direct effect on aggregate demand.

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The view that energy price shocks can have a demand-side impact is commonplace. Nevertheless, existing theoretical models do not capture a direct effect.¹ We posit two crucial features for this link. First, and in line with models that study the macroeconomic effects of energy price shocks, our small open-economy model features a constant elasticity of substitution (CES) production technology with low elasticity of substitution between labor and imported energy. The second key feature is household heterogeneity. We highlight the demand-side effects of this supply shock in a two-agent New Keynesian (TANK) model where agents differ in their sources of income and access to financial markets. The model features two types of households: *constrained hand-to-mouth (HtM) or worker households*, who consume only out of their labor income and have no access to financial markets, and *unconstrained households*, who earn firm profits in addition to labor income and have free access to financial markets.

We show that the impact of energy prices on aggregate demand depends critically on the elasticity of substitution between production inputs and household heterogeneity. The degree of substitutability among production inputs determines the response of households' income to the shock. Assuming production inputs are reasonably difficult to substitute, an increase in energy prices leads to a fall in the labor share of firms' expenditures. As households differ in their access to borrowing and sources of income, a reduction in the labor share adversely affects aggregate demand for two reasons. First, it implies a reduction in the flow of income accruing to domestic factors of production. Due to credit constraints faced by a share of households, this translates into lower demand. Second, as constrained worker households rely more heavily on labor income, a lower labor share implies a redistribution of income against agents with a high marginal propensity to consume, which further depresses aggregate demand. This supply shock therefore has a self-correcting effect, as the subsequent contraction in aggregate demand dampens inflationary pressures. To our knowledge, this is the first paper to study this transmission channel alongside its implications for optimal monetary policy.

The magnitude of this demand-side channel also depends on the degree of price rigidities, as the aforementioned contraction in aggregate demand can be moderated by the behavior of markups. Given price rigidities, an increase in energy prices reduces firms' markups. This redistributes income in favor of constrained worker households, hence increasing aggregate demand. Instead, with higher price flexibility, firms are able to pass the cost of the more expensive energy to workers by raising prices.

We show that an energy price shock is distinct among supply shocks in how it redistributes resources in the economy. An energy price shock has traditionally been proxied by an adverse technology shock, since both affect the productive capacity of the economy.² While both shocks raise firms' marginal costs, leading to an increase in inflation, they yield opposite effects on the demand side in our model. An adverse productivity shock leads to a fall in markups, as firms must hire more labor for the same amount of output. This increases constrained worker households' income, which boosts aggregate demand. However, an energy price shock lowers constrained worker households' income and leads to a fall in economic activity. Next, while an energy price shock and a markup shock both depress aggregate demand, the underlying cause is different. Higher markups imply an increase in the profit share relative to the labor share of income, redistributing resources against constrained worker households and depressing aggregate demand. The drop in demand is therefore fully explained by an uneven impact of the shock on households' income, due to the unequal income composition between constrained worker households and unconstrained firm-owning households. In contrast, the demand effect following an energy price shock is largely explained by a redistribution of resources towards the foreign sector, which affects demand due to households' unequal access to international credit markets.

The open-economy dimension is therefore crucial for explaining the dynamics in response to an energy price shock. As standard in the TANK literature, amplification in our model depends on the shock affecting constrained households by more relative to the unconstrained households. However, in our open-economy TANK model, the variable which captures the relative impact of the shock is the consumption gap, defined as the difference between unconstrained and constrained household consumption, rather than the income gap. These two variables differ since unconstrained worker households can smooth consumption by borrowing from abroad. The cyclical nature of the consumption gap therefore determines the amplification of shocks in an open-economy TANK model.

What is the optimal response of monetary policy to an energy price shock in our model and how does it depend on the degree of household heterogeneity? In contrast to a RANK economy, energy price shocks in the TANK economy have both supply and demand-side effects. While higher energy prices place upward pressure on inflation, they also constrain aggregate demand. Our analysis suggests that optimal monetary policy is contractionary in both RANK and TANK models to counteract the inflationary pressures. However, in the TANK model, the negative impact of higher energy prices on aggregate demand mitigates inflationary pressures. An energy price shock therefore has a weaker inflationary effect, which requires a milder increase in the interest rate. Finally, we explore conditions under which optimal policy may actually be expansionary in the presence of an adverse supply shock, finding this to be the case when the share of financially constrained worker households is sufficiently high.

To sharpen our results, we focus on energy solely as a production input in the baseline model. We therefore follow the conventional approach of modeling an energy price shock as a shock that constrains productive capacity, yet we demonstrate its direct effect on aggregate demand. Moreover, we find that an energy price shock can be regressive even though we abstract from features that would suggest such an effect. Specifically, we assume that both types of households are identical in labor supply and wages received.³ In an extension, we also consider the other extreme, where energy is solely a component of households' consumption baskets. Analogous to the baseline model, the effects of an energy price shock are contingent on the elasticity of substitution between energy and domestically produced goods in consumption. Due to complementarities, the higher cost of energy

¹ Models featuring an aggregate demand channel attribute the contraction in economic activity to the monetary policy response to inflation induced by energy price shocks (Medina and Soto, 2005).

² Bruno and Sachs (1985). See Kilian (2008) for references.

³ Känzig (2021b) shows that carbon taxation imposes a larger burden on low-income households, since they are disproportionately employed in demand-sensitive sectors.

leads to a reduction in the share of domestic goods in households' spending. As less resources are devoted to the purchase of domestically produced goods, households' income falls. While unconstrained worker households can maintain their consumption levels by borrowing from the foreign sector, constrained worker households must reduce their consumption, causing inequality to rise and aggregate demand to decline. We find that an energy price shock still leads to a contraction in aggregate demand, even when energy comprises an equal proportion of both constrained and unconstrained worker households' consumption baskets.⁴

The main difference between the scenarios with energy in production versus consumption is the impact of energy price shocks on markups and the speed of transmission to aggregate demand. When energy is an input for firms, costlier energy transmits only gradually to the price of consumption goods, resulting in a decrease in markups. Therefore, profits partially absorb the effects of costlier energy, limiting the impact of the shock on the constrained worker households. However, when energy enters directly into the consumption basket, markups no longer absorb the shock. The shock therefore affects constrained worker households directly and to a greater degree, which exacerbates inequality.

1.1. Related literature

Our paper contributes to a literature that studies the transmission of shocks in heterogeneous agent models (Furlanetto and Seneca, 2012; Galí et al., 2007) and the vast literature that studies the implications of household heterogeneity for macroeconomic dynamics more broadly (Galí et al., 2007; Debortoli and Galí, 2017; Auclert et al., 2021; Kaplan and Violante, 2018; Acharya and Dogra, 2020; Bilbiie, 2020; Broer et al., 2020; Cantore and Freund, 2021).

In concurrent and independent work, a closely related study by Auclert et al. (2023) employs a HANK model instead of a TANK model. Although HANK models can capture more realistic distributional effects and shock propagation, TANK models aim to preserve tractability while matching the key features of HANK models (Bilbiie, 2008; Debortoli and Galí, 2017; Cantore and Freund, 2021). This tractability facilitates our analysis of optimal monetary policy. In addition, the two agents in our model differ in access to credit as well as income type, which allows us to consider the unequal incidence of the energy price shock on labor income versus profit income. As a result, the behavior and evolution of markups is important in our model (alongside the degree of complementarity between energy and labor in production). We also show that an energy price shock is unique among supply shocks in terms of its demand-side impact, due to the way in which resources are redistributed between constrained worker households and unconstrained households.

This paper also contributes to the growing literature that examines the macroeconomic implications of the recent surge in imported energy prices (Bachmann et al., 2022; Gelain and Lorusso, 2022; Gornemann et al., 2022; Gagliardone and Gertler, 2023; Gnoco, 2023). Given the distributional effects of an energy price shock, a number of studies have examined the role of fiscal policy (Motyovszki, 2023; Kharroubi and Smets, 2023; Bayer et al., 2023).

Roadmap. The rest of this paper is structured as follows. In Section 2, we present the model with an emphasis on the key features: household heterogeneity and product input complementarity. Section 3 presents the baseline calibration and impulse response functions, which illustrate the transmission channels we discuss. We show how the magnitude of the various channels depend on the severity of credit constraints, price rigidities and the degree of substitutability between production inputs. We consider optimal monetary policy in Section 4. Section 5 explores an extension with energy as a consumption good. Section 6 concludes.

2. Baseline model

We begin our discussion of the baseline model with a focus on two key model features: household heterogeneity and imported energy as a complementary input to production.⁵

2.1. Household heterogeneity

Unconstrained households. A fraction $(1 - \omega)$ of households are financially *unconstrained* (denoted by u). They consume $C_{u,t}$, supply labor $N_{u,t}^h$ to unions, trade in domestic (foreign) nominal riskless bonds $B_{u,t}$ ($B_{u,t}^*$), and receive profits from firm ownership $DIV_{u,t}^F$. Their lifetime utility is given by

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left(\frac{C_{u,t}^{1-\sigma} - 1}{1-\sigma} - \chi \frac{(N_{u,t}^h)^{1+\varphi}}{1+\varphi} \right).$$

Unconstrained households maximize their lifetime utility subject to their budget constraint

$$W_t^h N_{u,t}^h + R_{t-1} B_{u,t-1} + \mathcal{E}_t \bar{R}^* B_{u,t-1}^* + DIV_{u,t}^F + DIV_{u,t}^L = P_t C_{u,t} + B_{u,t} + \mathcal{E}_t B_{u,t}^* + T_{u,t} + P_t \mathcal{T}_u, \quad (2.1)$$

where R_{t-1} ($\bar{R}^*$) denotes the gross nominal rate of return on domestic (foreign) bonds, P_t is the price of the consumption good, $\mathcal{E}_t$ is the nominal exchange rate (in domestic relative to foreign currency terms), $DIV_{u,t}^F$ represents profits derived from firm ownership, $DIV_{u,t}^L$ are profits transferred to the household by labor unions and $T_{u,t}$ are lump-sum transfers.⁶ $\mathcal{T}_u$ is a steady-state transfer from

⁴ Recent studies highlight the distributional impact of an energy price shock due to its effect on the consumption baskets of heterogeneous households (Bachmann et al., 2022; Peersman and Wauters, 2022; Bettarelli et al., 2023). Higher energy prices reduce purchasing power, particularly affecting poorer households, which allocate a larger portion of their income to energy expenses. This leads to a disproportionate inflationary burden on low-income households when energy prices rise.

⁵ In Section 5, we consider the case where energy is solely a consumption input. The full model derivation can be found in Appendix A.

⁶ We assume that households' consumption basket only consists of the domestically produced final output good.

unconstrained to constrained households. The unconstrained household's consumption-savings Euler equation is

$$1 = \mathbb{E}_t \left[\Lambda_{u,t,t+1} \frac{R_t}{\Pi_{t+1}} \right], \quad (2.2)$$

where $\Pi_t \equiv P_t/P_{t-1}$, $\Lambda_{u,t,t+1} \equiv \beta (C_{u,t}/C_{u,t+1})^\sigma$.

Constrained households. The remaining fraction ω of households are *financially constrained* (denoted by c) 'hand-to-mouth' households. They only receive labor income, hence their consumption is

$$P_t C_{c,t} = W_t^h N_{c,t}^h + DIV_{c,t}^L - T_{c,t} + P_t \mathcal{T}_c. \quad (2.3)$$

The wage received by households W_t^h is determined as a function of a weighted average of unconstrained and constrained households' marginal rate of substitution. Furthermore, we assume firms distribute labor demand equally among households, so that $N_{u,t}^h = N_{c,t}^h$. Aggregate consumption is

$$C_t = (1 - \omega)C_{u,t} + \omega C_{c,t}. \quad (2.4)$$

We define the consumption gap as the ratio between unconstrained and constrained consumption

$$\Gamma_t \equiv C_{u,t}/C_{c,t}. \quad (2.5)$$

2.2. Production input complementarity

Final output production involves perfectly competitive final good packers and monopolistically competitive final good producers.

Final good packers. Final good packers operate in a competitive market and produce the aggregate final good Z_t by combining a continuum of varieties $Z_t(i)$, $Z_t = \left(\int_0^1 (Z_t(i))^{\frac{\epsilon_z-1}{\epsilon_z}} di \right)^{\frac{\epsilon_z}{\epsilon_z-1}}$. Optimization implies the following demand function for variety i , $Z_t(i) = (P_t(i)/P_t)^{-\epsilon_z} Z_t$, where $P_t \equiv \left(\int_0^1 (P_t(i))^{1-\epsilon_z} di \right)^{\frac{1}{1-\epsilon_z}}$ is the price of the final composite good.

Final good producers. A continuum of final output producing firms, indexed by $i \in [0, 1]$, operate in a monopolistically competitive environment. Hence, each firm produces a single-differentiated good and operates as a monopoly in its own market. Firm i produces the final output variety $Z_t(i)$ using the following CES production technology with imported energy ($E_t^z(i)$) and labor ($N_t(i)$) as inputs

$$Z_t(i) = \varepsilon_t^{TFP} \left((1 - \alpha_{ez})^{\frac{1}{\psi_{ez}}} (N_t(i))^{\frac{\psi_{ez}-1}{\psi_{ez}}} + (\alpha_{ez})^{\frac{1}{\psi_{ez}}} (E_t^z(i))^{\frac{\psi_{ez}-1}{\psi_{ez}}} \right)^{\frac{\psi_{ez}}{\psi_{ez}-1}}, \quad (2.6)$$

where ε_t^{TFP} represents productivity and ψ_{ez} is the elasticity of substitution between energy and labor. The firm purchases labor from a union (paying the nominal wage W_t) and energy from an importer (at the nominal domestic currency price P_t^E).

Cost minimization. The final output producer's factor demand schedules are given by

$$W_t = (1 - \alpha_{ez})^{\frac{1}{\psi_{ez}}} \frac{MC_t^Z}{\tau_t^Z} \left(\frac{Z_t(i)}{N_t(i)} \right)^{\frac{1}{\psi_{ez}}} (\varepsilon_t^{TFP})^{\frac{\psi_{ez}-1}{\psi_{ez}}}, \quad P_t^E = (\alpha_{ez})^{\frac{1}{\psi_{ez}}} \frac{MC_t^Z}{\tau_t^Z} \left(\frac{Z_t(i)}{E_t^z(i)} \right)^{\frac{1}{\psi_{ez}}} (\varepsilon_t^{TFP})^{\frac{\psi_{ez}-1}{\psi_{ez}}},$$

where the Lagrange multiplier $MC_t^Z(i)$ is the (nominal) shadow cost of producing one more unit of final output, i.e. the nominal marginal cost, and $\tau_t^Z = \tau^Z \varepsilon_t^{\mathcal{M}_z}$ is a shock to final output marginal costs that is isomorphic to a price markup shock.

Price setting. Firms face price stickiness à la Calvo, resetting prices in every period with probability $(1 - \phi_z)$. A firm that is able to reset prices in period t chooses the price $P_t^\#$ that maximizes the sum of discounted profits subject to the demand faced in $t + s$

$$\mathbb{E}_t \sum_{s=0}^{\infty} (\phi_z)^s \{ \Lambda_{u,t,t+s} (P_t^\# Z_{t+s|t} - MC_{t+s}^Z Z_{t+s|t}) \} \quad s.t. \quad Z_{t+s|t} = \left(\frac{P_t^\#}{P_{t+s}} \right)^{-\epsilon_z} Z_{t+s}.$$

Profit maximization implies $\mathbb{E}_t \sum_{s=0}^{\infty} (\phi_z)^s \{ \Lambda_{u,t,t+s} Z_{t+s|t} (P_t^\# - \mathcal{M}_z MC_{t+s}^Z) \} = 0$, where $\mathcal{M}_z \equiv \frac{\epsilon_z}{\epsilon_z-1}$ is the desired final output price markup.

2.3. Remaining features

Wage stickiness. As described in detail in Appendix (A.2), we incorporate wage stickiness following [Schmitt-Grohe and Uribe \(2006\)](#). This gives rise to a standard wage inflation equation.

Imports. Each energy import good that the final output good producer demands is supplied by a perfectly competitive energy importer.⁷ Energy importers buy energy on the world market from foreign energy exporters at foreign currency energy price $P_t^{E,*}$ and sell it to domestic final output producers

$$p_t^E = Q_t p_t^{E,*}. \quad (2.7)$$

We define the real exchange rate $Q_t \equiv \mathcal{E}_t P_t^*/P_t$ and the real foreign and domestic energy prices are defined as $p_t^{E,*} \equiv P_t^{E,*}/P_t^*$ and $p_t^E \equiv P_t^E/P_t$. We model the foreign currency energy price level $p_t^{E,*}$ as an exogenous AR(1) process in which ε_t^E denotes an energy price shock process: $p_t^{E,*} = (p_{ss}^{E,*})^{1-\rho_E} (p_{t-1}^{E,*})^{\rho_E} \varepsilon_t^E$.

Exports. The global demand schedule for the bundle of domestic non-energy exports X_t depends on the foreign currency price of domestic non-energy exports, $P_t^{EXP} = P_t/\mathcal{E}_t$, relative to the world non-energy export price, P_{ss}^{X*} , and the world trade volume Y_{ss}^{*8}

$$X_t = \kappa^* \left(\frac{P_t^{EXP}}{P_{ss}^{X*}} \right)^{-\varsigma^*} Y_{ss}^*. \quad (2.8)$$

ς^* is the substitution elasticity between differentiated non-energy exports in the rest of the world.

Retailers. Perfectly competitive retailers buy final output goods from the final output packers at price P_t and convert them into domestic consumption goods and export goods.

Monetary policy. The central bank follows a Taylor rule that responds to deviations of (annual) inflation and the output gap from their targets,⁹

$$R_t = R^{1-\theta_R} R_{t-1}^{\theta_R} \left(\frac{\Pi_t^a}{\bar{\Pi}^a} \right)^{\frac{(1-\theta_R)\theta_\Pi}{4}} (\tilde{Y}_t)^{(1-\theta_R)\theta_Y}.$$

The output gap is defined as the ratio of employment to flexible price/wage employment, $\tilde{Y}_t = N_t/N_t^{flex}$.

Shock processes. The model features three shocks: (i) a TFP shock η_t^{TFP} (ii) a price markup shock $\eta_t^{M_z}$ and (iii) an energy price shock η_t^E

$$\begin{aligned} \log \varepsilon_t^{TFP} &= \rho_{TFP} \log \varepsilon_{t-1}^{TFP} - \varsigma_{TFP} \eta_t^{TFP}, & \eta_t^{TFP} &\sim N(0, 1) \\ \log \varepsilon_t^{M_z} &= \rho_{M_z} \log \varepsilon_{t-1}^{M_z} - \varsigma_{M_z} \eta_t^{M_z}, & \eta_t^{M_z} &\sim N(0, 1) \\ \log \varepsilon_t^E &= \varsigma_E \eta_t^E, & \eta_t^E &\sim N(0, 1) \end{aligned}$$

where ς_i , $i \in \{TFP, M_z, E\}$ denote the standard deviations of the shock processes.

2.4. Log-linearized model summary

Introducing lower case variables with a hat superscript to refer to log-deviations from steady state, we can summarize the key equations of the log-linearized model as follows.

Household demand. The aggregate consumption Euler equation (2.9) features the consumption gap $\hat{y}_t$. In the next section, we show how energy shocks affect aggregate demand through $\hat{y}_t$.

$$\hat{c}_t = \mathbf{E}_t[\hat{c}_{t+1}] + \mathbf{E}_t[\omega \Delta \hat{y}_{t+1}] - \frac{1}{\sigma} (\hat{r}_t - \mathbf{E}_t[\hat{r}_{t+1}]), \quad (2.9)$$

$$\hat{y}_t \equiv \hat{c}_{u,t} - \hat{c}_{c,t}, \quad \hat{c}_t = \omega \hat{c}_{c,t} + (1-\omega) \hat{c}_{u,t}. \quad (2.10)$$

Wage and price setting. Nominal wage and price stickiness gives rise to standard wage and price inflation equations (2.11) and (2.12). The log-linearized factor demand schedules are given by (2.13) and (2.14). Note that the time varying price markup is the inverse of the marginal cost, $\hat{\mu}_t^Z = -\hat{m}c_t^Z$.

$$\hat{\pi}_t^W = \frac{(1-\phi_w\beta)(1-\phi_w)}{\phi_w} (\hat{w}_t^h - \hat{w}_t) + \beta \mathbf{E}_t[\hat{\pi}_{t+1}^W], \quad \hat{\pi}_t^W = \hat{w}_t - \hat{w}_{t-1} + \hat{\pi}_t, \quad (2.11)$$

$$\hat{\pi}_t = \frac{(1-\phi_z\beta)(1-\phi_z)}{\phi_z} (\hat{m}c_t^Z) + \beta \mathbf{E}_t[\hat{\pi}_{t+1}], \quad (2.12)$$

$$\hat{m}c_t^Z = \hat{w}_t + \hat{\varepsilon}_t^{M_z} - \psi_{ez}^{-1} (\alpha_{ez} \hat{\varepsilon}_t^z - \alpha_{ez} \hat{n}_t) - \hat{\varepsilon}_t^{TFP}, \quad (2.13)$$

$$\hat{m}c_t^Z = \hat{p}_t^E + \hat{\varepsilon}_t^{M_z} - \psi_{ez}^{-1} ((1-\alpha_{ez}) \hat{n}_t - (1-\alpha_{ez}) \hat{\varepsilon}_t^z) - \hat{\varepsilon}_t^{TFP}. \quad (2.14)$$

⁷ We abstract from non-energy imports.

⁸ We assume that energy cannot be produced domestically and hence cannot be exported.

⁹ Note that in the case in which energy does not enter the consumption basket, domestic final output price 'core' inflation is equal to CPI inflation, $\Pi_t^a = \Pi_t^{CPI,a}$. See Section 5 for the case of energy solely as a consumption good.

Market clearing and monetary policy. Goods market clearing is given by Eq. (2.15) and implies that final output $\hat{z}_t$ is split between domestic consumption and exports

$$\hat{c}_t = \frac{Z_{ss}}{C_{ss}} \hat{z}_t - \frac{X_{ss}}{C_{ss}} \hat{x}_t, \quad \hat{z}_t = \hat{\varepsilon}_t^{TFP} + (1 - \alpha_{ez}) \hat{n}_t + \alpha_{ez} \hat{e}_t^z, \quad (2.15)$$

$$\hat{r}_t = \theta_R \hat{r}_{t-1} + (1 - \theta_R) \left((\theta_\pi/4) \hat{\pi}_t^{CPI,a} + \theta_Y (\hat{n}_t - \hat{n}_t^{lex}) \right), \quad \hat{\pi}_t^{CPI,a} \equiv \sum_{j=0}^3 \hat{\pi}_{t-j}^{CPI}, \quad \hat{\pi}_t^{CPI} = \hat{\pi}_t, \quad (2.16)$$

where the log-deviation of exports from steady state is $\hat{x}_t = \varsigma^* \hat{q}_t$, as implied by Eq. (2.8) and domestic energy prices are $\hat{p}_t^E = \hat{p}_t^{E,*} + \hat{q}_t$. As outlined above, $\hat{p}_t^{E,*}$ follows an exogenous AR(1) process and the real exchange rate $\hat{q}_t$ is pinned down by a UIP condition $\hat{q}_t = \mathbf{E}_t [\hat{q}_{t+1}] - (\hat{r}_t - \mathbf{E}_t [\hat{r}_{t+1}])$. Monetary policy sets the nominal interest rate $\hat{r}_t$ and follows a Taylor rule.

2.5. Demand-side effects of energy price shocks

Taking total employment $\hat{n}_t$ as a measure of domestic real activity and a proxy for value-added output (GDP), we can use (2.9), (2.10), (2.13), (2.14) and (2.15) to derive the following dynamic IS equation (see Appendix for details)

$$\begin{aligned} \hat{n}_t = & \mathbf{E}_t [\hat{n}_{t+1}] - \frac{1}{\sigma} \frac{C_{ss}}{Z_{ss}} (\hat{r}_t - \mathbf{E}_t [\hat{r}_{t+1}]) + \omega \frac{C_{ss}}{Z_{ss}} \mathbf{E}_t [\Delta \hat{y}_{t+1}] - \frac{X_{ss}}{Z_{ss}} \varsigma^* \mathbf{E}_t [\Delta \hat{q}_{t+1}] \\ & - \psi_{ez} \alpha_{ez} (\mathbf{E}_t [\Delta \hat{p}_{t+1}^E] - \mathbf{E}_t [\Delta \hat{w}_{t+1}]) + \mathbf{E}_t [\Delta \hat{\varepsilon}_{t+1}^{TFP}]. \end{aligned} \quad (2.17)$$

Solving this expression forward we obtain a dynamic IS curve that breaks down the channels through which energy prices affect activity

$$\begin{aligned} \hat{n}_t = & \underbrace{-\frac{1}{\sigma} \frac{C_{ss}}{Z_{ss}} \mathbf{E}_t \sum_{k=0}^{\infty} (\hat{r}_{t+k} - \hat{r}_{t+k+1})}_{\text{Inter-temporal substitution } (-)} \underbrace{-\omega \frac{C_{ss}}{Z_{ss}} \hat{y}_t}_{\text{Demand effect from credit constraints } (+/-)} \underbrace{+\psi_{ez} \alpha_{ez} (\hat{p}_t^E - \hat{w}_t)}_{\text{Intra-temporal substitution in production } (+)} \underbrace{+\frac{X_{ss}}{Z_{ss}} \varsigma^* \hat{q}_t}_{\text{terms of trade } (-)} - \hat{\varepsilon}_t^{TFP}. \end{aligned} \quad (2.18)$$

According to Eq. (2.18), economic activity $\hat{n}_t$ depends on the path of the real interest rate, the consumption gap $\hat{y}_t$, the price of energy relative to wages $\hat{p}_t^E - \hat{w}_t$, and on foreign demand for the domestic good, determined by the real exchange rate $\hat{q}_t$. An increase in the consumption gap $\hat{y}_t$ reflects redistribution against the constrained workers. As constrained agents have a higher marginal propensity to consume, such redistribution causes a drop in demand that results in a decrease in GDP. The effect of the consumption gap on GDP is increasing in the share ω of constrained households.

The IS equation illustrates the channels through which an energy price shock affects economic activity. Since our model nests a RANK ($\omega = 0$), the channels present in a RANK are also present here. In a RANK, energy prices operate through two different channels.

First, an increase in energy prices stimulates GDP through a higher price of energy $\hat{p}_t^E$ relative to wages $\hat{w}_t$, which leads to substitution from imported energy towards the domestic labor input (intra-temporal substitution effect).

Second, given the inflationary pressures derived from the shock, the central bank responds by tightening monetary policy. The ensuing increase in the real rate contracts economic activity (inter-temporal substitution effect) and leads to a fall in exports due to an exchange rate appreciation (terms-of-trade effect). This interest rate channel captures the usual mechanism through which supply shocks depress economic activity in an open-economy RANK model. In contrast to the TANK model, the energy shock itself is not contractionary in a RANK model. The economic downturn in the RANK model is purely a result of the monetary policy response to inflation caused by the energy price shock.

A new channel for supply shocks is present in the TANK economy, as indicated by the term involving the consumption gap. This term captures a demand-side impact of the energy shock that operates through an income effect. The sign of this demand-side effect depends on the response of the consumption gap to the shock. Under reasonable calibrations (where production inputs are largely complements), an increase in the price of energy translates into a contraction in households' income, as more resources must be devoted to the purchase of the energy input. Given financial constraints, demand by constrained worker households falls (as reflected by an increase in the consumption gap), leading to a downturn in economic activity.

Next, we discuss the aforementioned demand-side effect that emerges from credit constraints by analyzing how the energy price shock affects the consumption gap.

The consumption gap. Letting $INC_{u,t}$ and $INC_{c,t}$ denote unconstrained and constrained households' current income, and using the budget constraints (2.3) and (2.1), we decompose the consumption gap into an income gap $\Gamma_t^{inc} \equiv INC_{u,t}/INC_{c,t}$ and a foreign borrowing component¹⁰

$$\Gamma_t = \Gamma_t^{inc} + \frac{\mathcal{E}_t(\bar{R}^* - 1)B_{u,t-1}^* - \mathcal{E}_t \Delta B_{u,t}^*}{INC_{c,t}}. \quad (2.19)$$

¹⁰ The expression for the consumption gap takes into account that domestic bonds must equal zero in equilibrium.

Eq. (2.19) illustrates how an energy price shock affects the consumption gap through a differential impact on constrained and unconstrained households consumption. An unequal consumption response can have two sources. One source is through changes in the income gap, which reflects the different impact of the shock on current income (due to differences in income composition). The other source is unequal access to borrowing (reflected in changes in foreign bond holdings $\Delta B_{u,t}^*$), which allows unconstrained households to insure their consumption from income fluctuations following an energy price shock. From the economy's budget constraint, the evolution of foreign bonds depends on the balance of trade. Therefore, the consumption gap can be rewritten as

$$\Gamma_t = \Gamma_t^{inc} - \frac{1}{1-\omega} \frac{TB_t}{INC_{c,t}}, \quad (2.20)$$

where $TB_t = P_t^X X_t - P_t^E E_t^z$ is the trade balance. Using the definitions of unconstrained and constrained total income, the above expression can be written as follows

$$\Gamma_t = 1 + \underbrace{\frac{1}{1-\omega} \frac{\mathcal{M}_t^Z - 1}{\Xi_t^N}}_{\text{Income gap}} + \underbrace{\frac{1}{1-\omega} \left(\frac{1}{\Xi_t^N} - 1 - \frac{P_t^X X_t}{INC_{c,t}} \right)}_{\text{Borrowing}}, \quad \frac{\partial \Gamma_t}{\partial \mathcal{M}_t^Z} > 0, \quad \frac{\partial \Gamma_t}{\partial \Xi_t^N} < 0 \quad (2.21)$$

where $\mathcal{M}_t^Z \equiv P_t / MC_t^Z$ is firms' average markup and $\Xi_t^N \equiv W_t N_t / (W_t N_t + P_t^E E_t^z)$ is the labor share of firms' total expenditure. Eq. (2.21) shows that the effect of a change in energy prices on the consumption gap (and hence, on demand) is determined by the impact of the shock on two key variables, firms' markups and the labor share.

Firms' markups positively affect the income gap, since an increase in the markup redistributes resources towards unconstrained firm owners. However, the labor share negatively affects both the borrowing and income gap components. A decrease in the labor share diminishes income for both unconstrained and constrained households. While the former can mitigate this decline by borrowing to maintain consumption levels, the latter must cut demand, thereby increasing the consumption gap, as reflected by an increase in the borrowing term.¹¹ A reduction in the labor share also increases the income gap, since constrained households disproportionately depend on labor income.

Finally, to understand how the energy price shock affects firms' average markup and the labor share, notice that these two objects are linked to the price of energy according to the following expressions¹²

$$\mathcal{M}_t^Z = \frac{\epsilon_t^{TFP} P_t}{\left((1-\alpha_{ez}) W_t^{1-\psi_{ez}} + \alpha_{ez} (P_t^E)^{1-\psi_{ez}} \right)^{\frac{1}{1-\psi_{ez}}}}, \quad \frac{\partial \mathcal{M}_t^Z}{\partial P_t^E} < 0 \quad (2.22)$$

$$\Xi_t^N = \left(1 + \frac{\alpha_{ez}}{1-\alpha_{ez}} \left(\frac{P_t^E}{W_t} \right)^{1-\psi_{ez}} \right)^{-1}, \quad \frac{\partial \Xi_t^N}{\partial P_t^E} < 0. \quad (2.23)$$

Notice from (2.22) that, given price rigidities, an increase in energy prices (P_t^E) reduces firms' markups. This implies a redistribution of income in favor of workers, resulting in a reduction of the consumption gap (see Eq. (2.21)). This stimulates aggregate demand, and hence, economic activity (Eq. (2.18)). Eq. (2.23) shows that the impact of higher energy prices on the labor share depends crucially on the elasticity of substitution between energy and labor (ψ_{ez}). In the case of a Cobb–Douglas production technology ($\psi_{ez} = 1$) we have $\Xi_t^N = 1 - \alpha_{ez}$, implying that the price of energy has no impact on the labor share. If the elasticity of substitution is larger than one ($\psi_{ez} > 1$), higher energy prices increase the labor share. The reason is that costlier energy triggers a strong substitution from energy towards labor. The resulting redistribution of income in favor of domestic households boosts constrained agents consumption, leading to a decrease in the consumption gap. As a result, aggregate demand and economic activity increase. Alternatively, if energy cannot easily be substituted for by labor ($\psi_{ez} < 1$), an increase in energy prices reduces the labor share. The shock therefore redistributes against the domestic households, increases the consumption gap due to a decrease in constrained agents consumption and depresses economic activity.

In the next section, we cite empirical evidence that finds low substitutability between labor and energy. In this scenario, we should expect that an increase in energy prices will reduce both the labor share and firms' markups (i.e., the profit share). The relative effect of the shock on these two objects will determine the extent of its impact on constrained agents' income, and hence, the size of the demand-side effect of the energy price shock.

3. Dynamic responses under a Taylor rule

Parameterization. To align with existing literature, we calibrate our model using commonly employed parameterizations. We assume a discount factor, β , of 0.9994. The elasticities of substitution across goods varieties (ϵ_z) and across worker types (ϵ_w) are both set to 11, which implies a markup of 10% in steady state. We assume goods prices and wages are adjusted with Calvo parameters $\phi_z = 0.66$ and $\phi_w = 0.92$, respectively. We set the response to inflation (θ_π) and slack (θ_Y) in the Taylor rule to 1.5 and 0.125,

¹¹ Notice that the reduction in the labor share reflects an increase in the resources devoted to imported energy (an increase in the energy share), which must be financed via an increase in foreign debt. Borrowing from the foreign sector is used by unconstrained households to finance their consumption.

¹² The expression for the labor share is obtained using firms' demand functions for energy and labor.

respectively.¹³ The interest rate smoothing parameter (θ_R) is set to 0.9. The productivity process parameters are set to $\rho_{TFP} = 0.93$ (persistence) and $\zeta_{TFP} = 0.07$ (standard deviation). The energy price shock has persistence $\rho_E = 0.8$ and standard deviation $\zeta_E = 1$. In the experiments below, we will look at the case of an increase of energy prices by 100% on impact. The markup shock has persistence $\rho_{M_z} = 0.9$ and standard deviation $\zeta_{M_z} = 0.1$. The population share of constrained worker households (ω) is set to 0.25. A calibration of the key model parameters is available in the Appendix.

We limit our discussion to the following key parameters: the elasticity of substitution between energy and labor in production ($\psi_{ez} = 0.15$)¹⁴ and the steady state share of energy in production ($\alpha_{ez} = 0.05$).¹⁵ There are a wide range of estimates for the elasticity of substitution between production inputs in the literature. Higher estimates, such as those provided by [Bodenstein et al. \(2012\)](#) (0.42) are motivated by measures of the short-run price elasticity of oil demand from structural econometric models. [Natal \(2012\)](#) sets this parameter to 0.3, while [Plante \(2014\)](#) suggests a calibration of 0.25 so that the own price elasticity of oil is approximately -0.25 . Lower values of around 0.2 can be found in [Krichene \(2002\)](#) and [Montoro \(2012\)](#), who set the value of the elasticity of substitution between oil and labor at 0.2, equal to the average value reported by [Hamilton \(2009\)](#). Finally, [Stevens \(2015\)](#) suggests an elasticity of substitution between oil and value-added of 0.03.¹⁶ Regarding the share of energy in production, parameterizations range from 2% in [Natal \(2012\)](#) for the US, 4% in [Bachmann et al. \(2022\)](#) for Germany, and 5% in [Stevens \(2015\)](#).

Energy price shock IRFs. In [Fig. 1](#), we show the response to an increase in energy prices in the baseline model outlined above. In the RANK economy (red lines), an energy price shock places upward pressure on production costs, leading to a surge in inflation. The central bank responds by tightening monetary policy, which induces a contraction in activity. Relative to the RANK, the TANK economy (blue lines) experiences a deeper contraction. Moreover, while the recession in the RANK originates from the contractionary policy implemented by the central bank, in the TANK, it is driven to a considerable degree by the direct impact of higher energy prices on aggregate demand. Since production inputs are complementary in the TANK economy, higher energy prices reduce the labor share of total income, implying a drop in labor earnings.¹⁷ Given borrowing constraints for worker households, this translates into a fall in aggregate demand. The employment decomposition in [Fig. 1](#) shows how much the fall in demand as a result of credit frictions contributes to the contraction in employment in the TANK economy. Due to the adverse effect of the energy price shock on demand, monetary policy in the TANK is looser. More accommodative monetary policy limits the decline in employment in the TANK economy compared to the RANK economy. In the TANK, the more accommodative monetary policy counteracts the contractionary impact of energy prices resulting from credit constraints. If simulations were conditioned on the RANK-implied policy path of the real interest rate, the TANK economy would experience a deeper contraction in employment compared to the RANK. In the first row and fourth column panel of [Fig. 1](#) we show the dynamics of the consumption gap and its drivers (see Eq. (2.21)). The consumption gap fluctuates due to households' unequal income composition (green square line) and unequal access to foreign borrowing (pink triangle line). While consumption of constrained worker households largely falls due to the drop in their income, unconstrained firm owner households are able to insure their consumption by borrowing from the external sector. The unequal access to borrowing (pink triangle line) explains the increase in the consumption gap. Initially, a drop in firms' markups (due to costlier energy) results in a negative income gap, which limits the increase in the consumption gap. Over time, as firms pass the costs of the shock to constrained HtM households through an increase in prices, markups recover and the income gap goes up as well, further raising the gap in consumption.¹⁸

Interaction between an open economy environment and borrowing constraints. Two key features of our model interact to generate a demand side effect: an open economy setting, where energy is imported, and credit constraints. In this context, higher energy prices result in a deterioration of the terms of trade, leading to income redistribution favoring the foreign sector over domestic households. If households had unrestricted access to savings and credit, decreases in income would not substantially influence their spending decisions. However, if we assume that certain households face limitations in accessing credit, reductions in income can significantly impact demand. Consequently, the demand-side effect arises from the interaction between the open economy setting with imported energy and credit constraints.

Role of complementarities. [Fig. 2](#) illustrates the effects of the energy price shock under a higher degree of substitutability between labor and energy in production (Cobb–Douglas, $\psi_{ez} = 1$). The increase in the price of energy imports implies a fall in the relative price of labor, which now leads to greater degree of substitution towards labor. The labor demand schedule shifts upwards and employment increases. For $\psi_{ez} = 1$, higher employment fully compensates for the lower relative wage, leaving the labor share

¹³ As highlighted by [Schmitt-Grohe and Uribe \(2006\)](#), within a Calvo setting, two alternative modeling assumptions can be considered regarding household labor input. First, households act as monopolistic suppliers of differentiated labor inputs, with hours worked varying across households. Alternatively, as in our model, households provide a uniform labor input, which is then transformed into a diversified labor input by unions, with households providing equal hours. In the former case, empirical estimates of the wage Phillips curve slope suggest a Calvo parameterization associated with a wage duration of 3 to 4 quarters. In contrast, in the latter case, the Calvo parameter must be calibrated to imply a wage duration between 10 to 12 quarters to align with empirical slope estimates.

¹⁴ In the Appendix, in [Figure B.16](#) we also consider ($\psi_{ez} = 0.3$) and show that the IRFs for ($\psi_{ez} = 0.15$) and ($\psi_{ez} = 0.3$) are very similar.

¹⁵ In [Section 5](#) we will introduce an extension of the model in which we allow for energy in the consumption basket. For simplicity, we will chose the same energy share and the same degree of substitutability as discussed here, $\alpha_{ec} = 0.05$ and $\psi_{ec} = 0.15$.

¹⁶ This parameter is equivalent to the short-run oil demand elasticity and is chosen to be consistent with reduced-form evidence on the slope of the oil demand curve that lies between 0 and 0.11.

¹⁷ In [Appendix C](#), we use data from the UK's Office for National Statistics' supply-use tables to show a decrease in the labor share in firms' expenditure following episodes of heightened energy prices, which suggests limited substitutability between energy and labor inputs in production.

¹⁸ The income gap eventually goes up due to the decline in the labor share, as constrained worker households rely disproportionately on labor income.

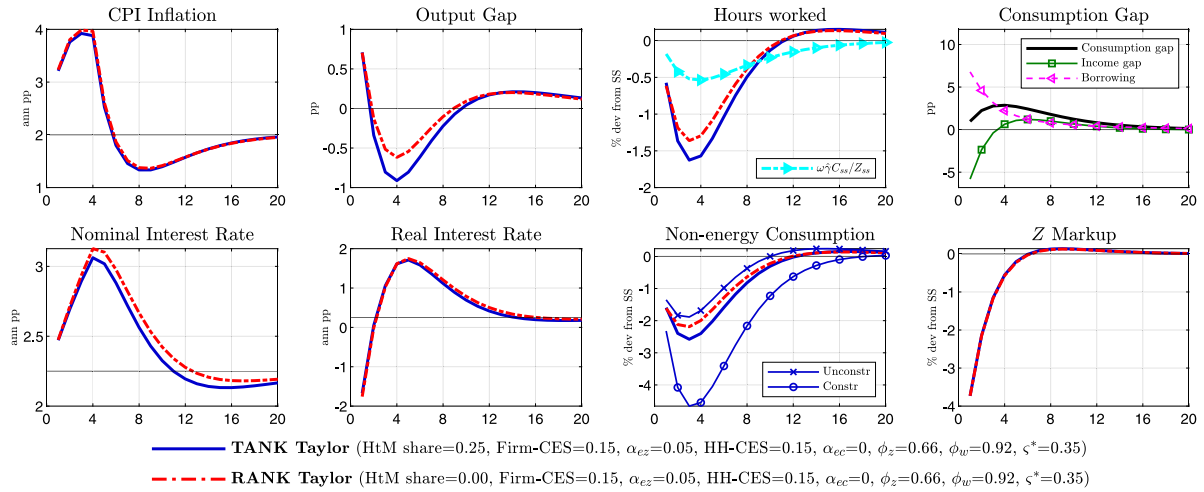


Fig. 1. Dynamic responses to a global energy price shock: Taylor policy for TANK vs. RANK. *Notes:* This figure shows the IRFs of key model variables to a 100% increase in the foreign currency price of energy. For hours worked, the light blue dashed line captures $\hat{\gamma}_t$ -term in Eq. (2.18), which represents the variation due to the demand effect from credit constraints. Full set in Figure B.2 in the Appendix.

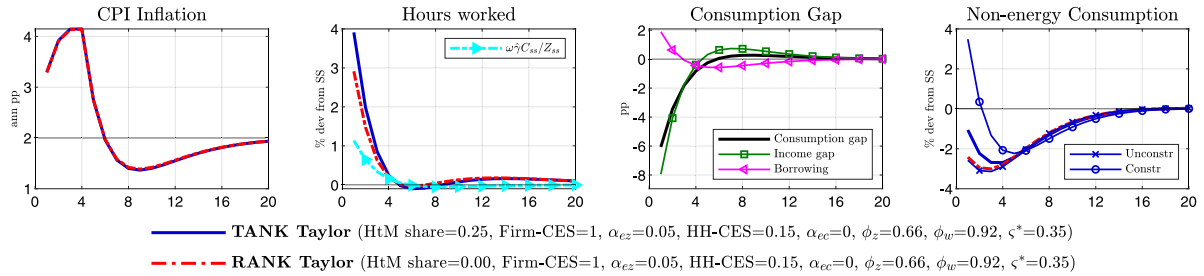


Fig. 2. Dynamic responses to a global energy price shock: Cobb-Douglas instead of CES production function. *Notes:* This figure shows the IRFs of key model variables to a 100% increase in the foreign currency price of energy. For hours worked, the light blue dashed line captures $\hat{\gamma}_t$ -term in Eq. (2.18), which represents the variation due to the demand effect from credit constraints. Full set in Figure B.17 in the Appendix.

constant. While the labor share remains constant, firms' markups experience a reduction due to higher marginal costs. Redistribution in favor of constrained worker households, reflected in a reduction of the consumption gap, boosts aggregate demand. Given the positive effect of the shock on demand, the TANK economy experiences a milder recession relative to its RANK counterpart.

In Appendix D, we provide empirical IRFs to illustrate a decline in hours following an energy shock, consistent with our simulations based on the baseline calibration ($\psi_{ez} = 0.15$), and in contrast to simulations assuming a unitary elasticity of substitution ($\psi_{ez} = 1$).

Higher share of constrained households. Fig. 3 illustrates the effects of the energy price shock when we assume a larger share of constrained households. Given more severe borrowing constraints, consumption becomes more responsive to the drop in households' income. It follows that the increase in energy prices induces a stronger fall in aggregate demand, leading to a deeper recession than in the baseline case (Fig. 1).

In Appendix B, we provide additional robustness checks of our results by considering alternative calibrations of the Calvo price and wage parameters, as well as variations in the share of constrained agents and the degree of substitutability between labor and energy.

3.1. The demand-side effects of alternate supply shocks

We consider the dynamics after a productivity shock to demonstrate that an energy price shock is distinct from other supply shocks in how it redistributes resources in the economy.

TFP shock IRFs. An adverse TFP shock is often used to proxy the effects of an energy price shock, since both types of shocks constrain supply. To analyze the demand-side effect of a shock to firms' TFP, consider Eqs. (2.21) to (2.23). With lower productivity, firms must hire more labor to produce each unit of output. This implies lower markups (Eq. (2.22)) and an increase in labor income, out of which constrained hand-to-mouth households have a high propensity to consume. It follows that the consumption gap falls and

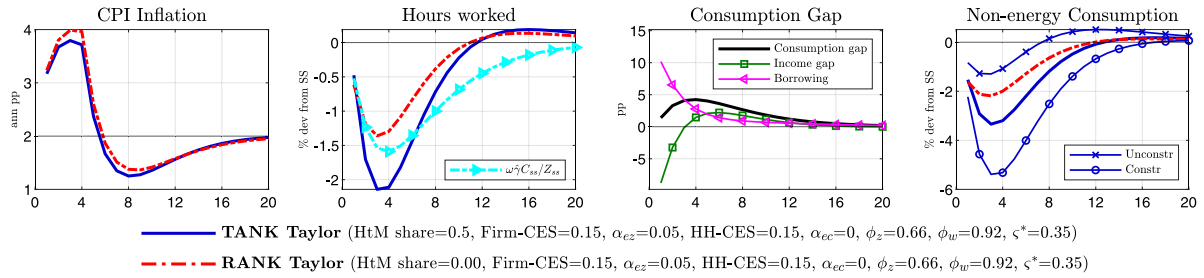


Fig. 3. Dynamic responses to a global energy price shock: Higher share of constrained households. *Notes:* This figure shows the IRFs of key model variables to a 100% increase in the foreign currency price of energy. For hours worked, the light blue dashed line captures $\hat{\gamma}_t$ -term in Eq. (2.18), which represents the variation due to the demand effect from credit constraints.

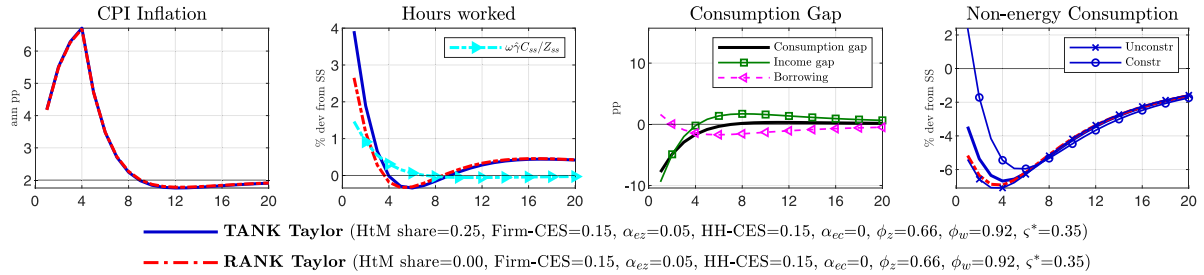


Fig. 4. Dynamic responses to a TFP shock: Taylor policy for TANK vs. RANK. *Notes:* This figure shows the IRFs of key model variables to a 7% drop in TFP. For hours worked, the light blue dashed line captures $\hat{\gamma}_t$ -term in Eq. (2.18), which represents the variation due to the demand effect from credit constraints.

GDP increases (Eq. (2.18)). The IRFs to an adverse TFP shock in Fig. 4 illustrate this intuition. Similar to the energy price shock, the TFP shock leads to higher marginal costs, which places upward pressure on inflation. The response of the central bank to higher inflation leads to a drop in output. While both energy and TFP shocks generate similar supply-side effects, this is not the case for the demand-side effect. Lower TFP implies that *more* labor is required to produce each unit of the good, which explains the observed increase in employment. Workers' income thus increases, boosting aggregate demand. As a consequence, the TANK economy (blue lines) features a milder contraction in consumption and output relative to the RANK economy (red lines). Energy and TFP shocks therefore diverge in terms of their impact on demand. Whereas the former reduces workers' income, the latter increases it, leading to a different profile for aggregate demand.¹⁹

Figure B.1 in the Appendix shows the IRFs after a markup shock. Like the energy price shock, a markup shock raises the consumption gap, hence depressing aggregate demand. However, in the case of a markup shock, the rise in the consumption gap is fully explained by households' unequal income composition. Instead, with the energy price shock, the rise in the consumption gap is largely explained by unequal access to international credit markets.

4. Optimal monetary policy

To compute the Ramsey-optimal policy, we assume a utilitarian policymaker that attaches equal weights to the utility of all households, maximizing their lifetime utility subject to the non-linear system of equations that describe private agents' optimality conditions. Fig. 5 presents the IRFs under the Ramsey policy. We compare the optimal policy in the TANK (blue lines) versus the RANK model (red lines). Although optimal policy leads to very similar paths for inflation and employment in the two economies, the implementation is different. In both cases, the policymaker implements contractionary policy in order to counteract the inflationary effect of the shock. However, the optimal increase in the path of the real interest rate is milder in the TANK model than in the RANK model. This is explained by the direct contractionary effect of higher energy prices on households' income. In the TANK model, lower income translates into lower aggregate demand, which contains the inflationary pressures of the shock. Hence, a less aggressive tightening of policy is optimal.²⁰

A key difference between the Taylor-rule IRFs and the Ramsey-optimal IRFs is the response of wage inflation (see Figures B.2 and B.3 in the Appendix). Under Ramsey-policy, wage inflation is nearly perfectly stabilized. This is because, in settings with sticky prices and wages, central banks care about the stabilization of both price and wage inflation. The central bank's tolerance

¹⁹ De Giorgi and Gambetti (2017) provide empirical evidence of pro-cyclical consumption inequality in response to a TFP shock.

²⁰ The initial impact of the nominal interest rate is more pronounced in the TANK model compared to the RANK model, and subsequently, it is reduced more swiftly. This rise in the nominal interest rate at the outset counteracts the expansionary effect on current consumption resulting from looser policy implemented further out.

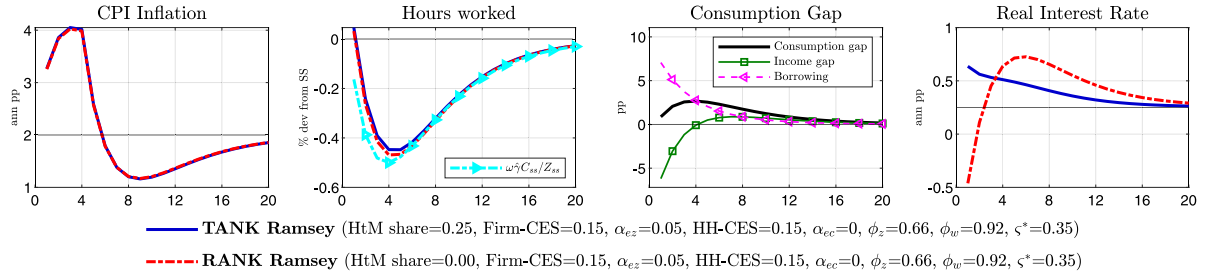


Fig. 5. Dynamic responses to a global energy price shock: Ramsey-policy for TANK vs. RANK. Notes: This figure shows the IRFs of key model variables to a 100% increase in the foreign currency price of energy. For hours worked, the light blue dashed line captures $\hat{\gamma}_t$ -term in Eq. (2.18), which represents the variation due to the demand effect from credit constraints. Full set in Figure B.3 in the Appendix.

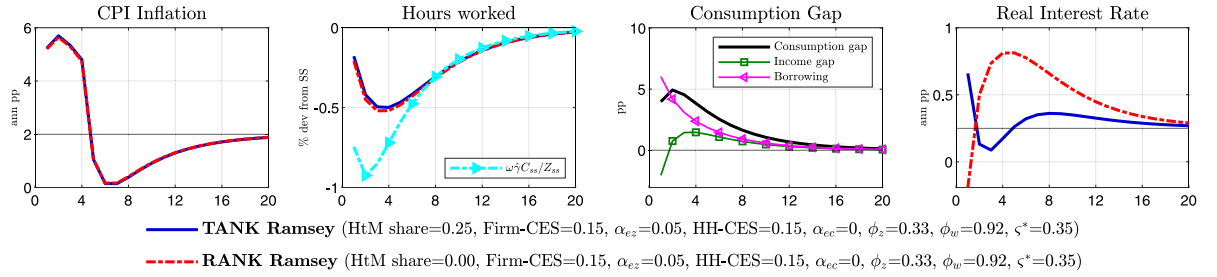


Fig. 6. Dynamic responses to a global energy price shock: Ramsey-policy with higher price flexibility. Notes: This figure shows the IRFs to a 100% increase in the foreign currency price of energy. For hours worked, the light blue dashed line captures $\hat{\gamma}_t$ -term in Eq. (2.18), which represents the variation due to the demand effect from credit constraints. Full set of IRFs in Figure B.13 in the Appendix.

of higher price inflation volatility relative to wages may be linked to the assumed degree of wage stickiness in relation to prices.²¹ Additionally, it is noteworthy that in TANK models, the central bank must take into account inequality in the design of optimal policy.²² However, our simulations indicate nearly identical inflation and activity paths under RANK and TANK, suggesting that inequality considerations have a minimal impact on optimal policy in the TANK economy following an energy price shock. This aligns with findings by [Debortoli and Gali \(2017\)](#), who conclude that after a productivity shock, optimal policy recommendations in a TANK model do not significantly differ from those in a RANK model.²³ They attribute this result to the low weight attached to consumption inequality volatility in the loss function.²⁴

Role of price stickiness. As stressed earlier, the effect of higher energy prices on aggregate demand depends on the evolution of firms' markups. If price inflation remains contained despite costlier energy, firms largely absorb the costs of the shock. This would be reflected in a reduction in markups. However, if prices go up strongly to preserve markups, firms can pass the costs of the shock to workers, who will experience a more severe reduction in their income. The degree to which prices react to the shock thus determines who bears the cost of an energy price shock, and hence, its impact on aggregate demand. In [Fig. 6](#) we consider a scenario where firms raise prices more aggressively in response to the costlier energy input in an attempt to preserve profits. Precisely, we assume a higher degree of price flexibility, setting the Calvo pricing parameter to $\phi_z = 0.33$. Comparing [Fig. 6](#) with [Fig. 5](#) illustrates that when firms respond to an energy price shock by significantly raising prices, constrained worker households experience a more substantial decline in income relative to unconstrained households, as indicated by the income gap. Since the constrained households are more severely affected, there is a deeper contraction in aggregate demand. As a consequence, optimal monetary policy in the TANK model is now much looser relative to its RANK counterpart.

Role of credit constraints. Next, we explore whether optimal policy can actually be expansionary in response to an adverse supply shock. As the contractionary impact of the shock on demand intensifies, it becomes optimal for policymakers to ease policy. To measure the policy stance, we introduce a metric indicating whether policy is contractionary or expansionary. From (2.2), we know that the demand of unconstrained households is determined by the expected and cumulated path of the real interest rate, rather than the current spot real rate. Therefore, we define the *policy stance* as $\hat{s}_t \equiv \sigma^{-1} \mathbb{E}_t \sum_{k=0}^{\infty} (\hat{r}_{t+k} - \hat{\pi}_{t+k+1})$.

²¹ [Erceg et al. \(2000\)](#) shows that in settings characterized by sticky prices and wages, central banks must aim to stabilize wage inflation alongside price inflation. Additionally, central banks should tolerate relatively higher volatility in prices compared to wages if prices exhibit greater flexibility than wages.

²² [Bilbiie and Ragot \(2021\)](#) and [Debortoli and Gali \(2017\)](#) find that welfare losses are dependent on the volatility of a measure of consumption inequality in TANK models.

²³ See also [Ascari et al. \(2017\)](#).

²⁴ Refer to Appendix, Section A.10 for more details.

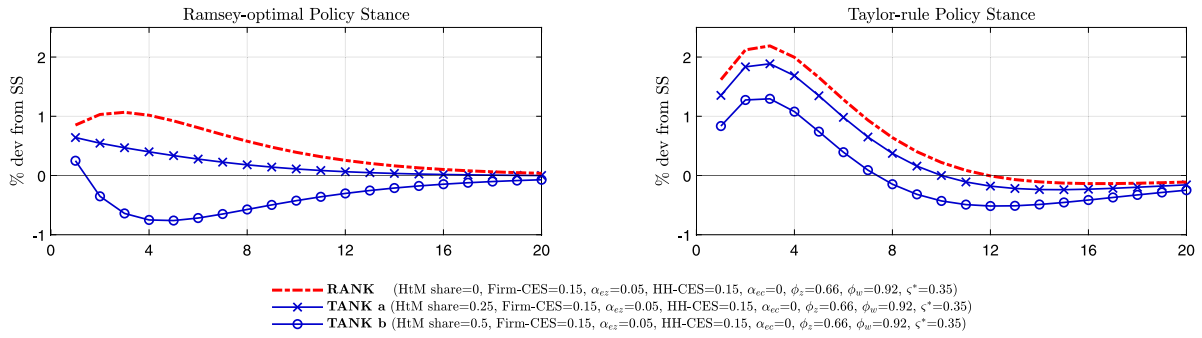


Fig. 7. Dynamic responses to a global energy price shock: Policy with stronger credit constraints. *Notes:* This figure shows the IRFs of the policy stance to a 100% increase in the foreign currency price of energy. In the panel on the left (right) the central bank implements Ramsey-optimal policy (follows a Taylor rule). Full set in Figure B.4 in the Appendix.

Fig. 7 presents the IRFs for the policy stance over an increasingly larger share of constrained agents, which allows the energy price shock to yield a correspondingly larger fall in households' consumption. The left panel depicts the case of Ramsey-optimal policy, while the right panel depicts the case of a Taylor rule. In the RANK model, Ramsey-optimal monetary policy remains contractionary throughout the period of higher energy prices in order to counteract inflation. Meanwhile, in the TANK, optimal policy under a constrained household share of $\omega = 0.25$ is less contractionary. Under a higher share of constrained households, $\omega = 0.5$, the optimal policy stance can even be accommodative.

5. Extension: Energy as a consumption good

In this section, we extend our model to incorporate imported energy as a component of households' consumption baskets. Household consumption is given by the following CES aggregator

$$CES_{j,t} = \left((1 - \alpha_{ec})^{\frac{1}{\psi_{ec}}} (C_{j,t})^{\frac{\psi_{ec}-1}{\psi_{ec}}} + \alpha_{ec}^{\frac{1}{\psi_{ec}}} (E_{j,t}^h)^{\frac{\psi_{ec}-1}{\psi_{ec}}} \right)^{\frac{\psi_{ec}}{\psi_{ec}-1}} \quad \forall j \in \{u, c\}$$

where $C_{u,t}$ and $E_{u,t}^h$ are consumption of the domestic good and energy for unconstrained households, and α_{ec} denotes the share of energy in expenditure. The analogous variables with subscript c denote the counterparts for constrained agents. Parameter ψ_{ec} is the elasticity of substitution between the domestic good and energy.

The consumption gap. With energy in the consumption basket, the consumption gap (the ratio of the unconstrained relative to the constrained household consumption bundle) depends positively on firms' average markup ($\hat{\mu}_t^Z$) and the price of energy (p_t^E), as illustrated by the following log-linear expression

$$\Gamma \hat{\gamma}_t = \underbrace{\frac{1}{1-\omega} M_1 \mathcal{M} \hat{\mu}_t^Z}_{\text{Income gap}} + \underbrace{\frac{\mathcal{M}}{1-\omega} \left(B_2 (\widehat{ces}_t - \hat{x}_t) + B_3 \hat{\mu}_t^Z + B_4 \left(1 - \frac{C}{Z} \psi_{ec} \right) \hat{p}_t^E \right)}_{\text{Borrowing}},$$

where $M_1 > 1$, $B_2 > 0$, $B_3 < 0$, $B_4 > 0$ are auxiliary terms.²⁵ We assume that energy does not enter the production function in order to focus on its effects in the consumption basket.

An increase in firms' markup increases the consumption gap ($\partial \hat{\gamma}_t / \partial \hat{\mu}_t^Z > 0$).²⁶ Note, however, that if energy only enters in the consumption basket, then the price of energy would have no direct effect on markups ($\partial \hat{\mu}_t^Z / \partial \hat{p}_t^E = 0$). This is a key difference relative to the model with energy as a production input (Eq. (2.22)). In that case, the higher price of energy gradually passed through to the price of the consumption good. Therefore, firms would partially absorb the impact of the costlier energy through a fall in markups. However, with energy in the consumption basket, energy prices instantaneously pass through to the price of the consumption good. Without a fall in markups to absorb the shock, constrained households are more strongly affected by higher energy prices.

An increase in energy prices widens the consumption gap ($\partial \hat{\gamma}_t / \partial \hat{p}_t^E > 0$) if the substitutability between energy and non-energy goods in consumption is low enough, $\psi_{ec} < Z/C$.²⁷ The intuition is that the share of domestic goods in households' expenditure responds to changes in the price of energy, and the direction of this response depends on the elasticity of substitution between the domestic good and energy (ψ_{ec}). With a low elasticity of substitution, less resources are spent on the purchase of domestic products following an increase in energy prices, leading to a drop in households' income. While unconstrained households are able to maintain their consumption levels by borrowing from the foreign sector, constrained worker households must cut demand. The consumption gap therefore increases and aggregate demand falls.

²⁵ $M_1 \equiv \frac{wN+E^Z}{wN} > 1$, $B_2 \equiv M_1 \frac{CES}{Z} \frac{X}{Z} > 0$, $B_3 \equiv M_1 \left(\frac{CES}{Z} - 1 \right) < 0$, $B_4 \equiv M_1 \frac{E^h}{Z} > 0$ Note that $CES/Z = (1 - X/Z) = 0.75$ in our calibration.

²⁶ Note that $\partial \hat{\gamma}_t / \partial \hat{\mu}_t^Z = \mathcal{M} M_1 / (1 - \omega) (CES/Z) > 0$.

²⁷ The threshold is $Z/C = 1/(1 - 0.25) = 1.333$ in our calibration.

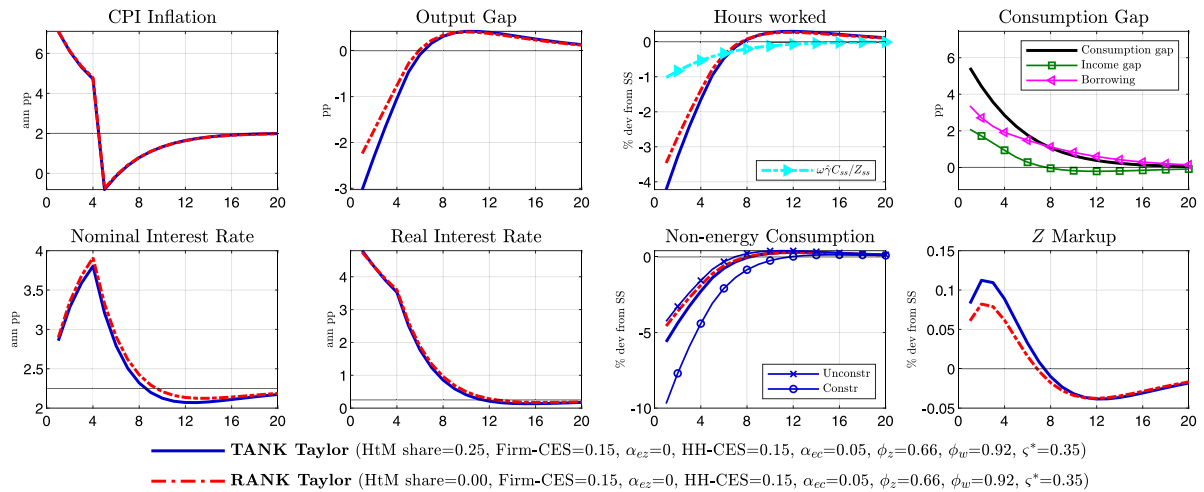


Fig. 8. Dynamic responses to a global energy price shock: Energy only as a consumption good. *Notes:* This figure shows the IRFs of key model variables to a 100% increase in the foreign currency price of energy. For hours worked, the light blue dashed line captures $\hat{\gamma}_t$ -term in Eq. (2.18), which represents the variation due to the demand effect from credit constraints. Full set in Figure B.5 in the Appendix.

Impulse response functions. Fig. 8 shows the response to an energy price shock when energy is a component of households' consumption basket.²⁸ Households react to costlier energy by substituting it with the domestically produced good. However, given a low elasticity of substitution, the share of domestically produced goods in total expenditure decreases. Consequently, domestic households' income falls. This leads to a decline in constrained agents' consumption, resulting in a larger consumption gap and lower aggregate demand. As a result, the TANK economy experiences a more severe contraction than the RANK. Compared to the scenario where energy is only a production input (Fig. 1), there is a larger impact on consumption inequality. This is due to the income gap, which now increases. The different evolution of the income gap follows from the different response of firms' markups. With energy as an input, markups fall, partially absorbing the shock. With energy entering directly in households' consumption basket, markups do not attenuate the impact of the shock on inequality.

6. Conclusion

We build an open economy model with household heterogeneity and complementarity between imported energy and labor in production to highlight the demand-side effects of an energy price shock. We show that an energy price shock has different effects on households, depending on their sources of income and borrowing constraints. Due to complementarities between imported energy and labor in production, an energy price shock leads to a fall in the labor share of total factor expenditure, reducing the flow of income accruing to domestic inputs, which depresses aggregate demand. Given borrowing constraints, the fall in aggregate demand is stronger in the TANK model, compared to a RANK model. We find similar results in an extension with energy in the consumption basket. The magnitude of this demand-side effect depends on the elasticity of substitution between production inputs, the degree of price flexibility, and the share of constrained households. We also show that this effect is not a general feature of supply shocks, as an energy price shock is unique in how it redistributes resources in the economy. The adverse effect of an energy price shock on aggregate demand in our TANK model implies that Ramsey-optimal monetary policy is less contractionary relative to a RANK model. When credit constraints are severe, optimal monetary policy may even be expansionary.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.jmoneco.2024.103620>.

²⁸ Two additional parameters are needed relative to the baseline model calibration in Table 1 in the Appendix: the proportion of energy in the consumption basket of households $\alpha_{ec} = 0.05$ and the elasticity of substitution between the domestic good and energy $\psi_{ec} = 0.15$.

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